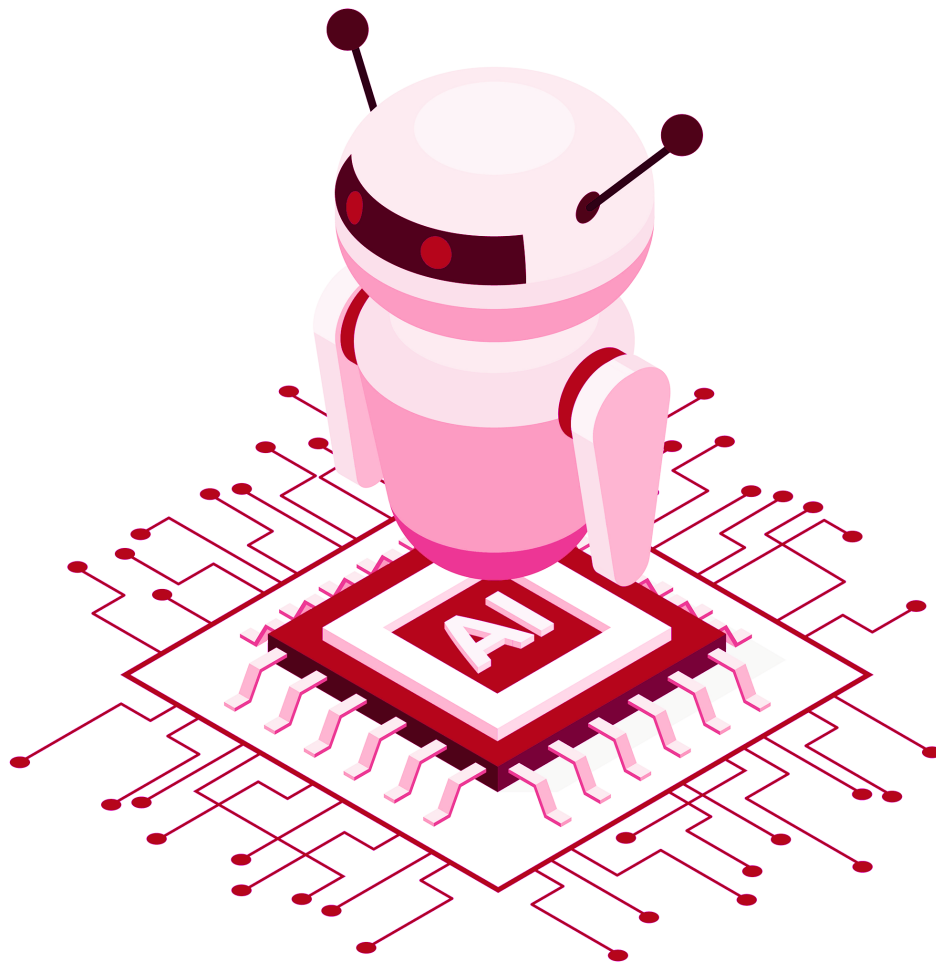




Deep Reinforcement Learning

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Homework 7:

Value-Based Theory

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Grading

The grading will be based on the following criteria, with a total of 100 points:

Section	Points
Positive Rewards	15
General Rewards	10
Policy Turn	25
Bellman Operators	15
Bellman Residuals	35
Bonus 1: Writing your report in Latex	5
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1 Iteration Family

Let $M = (S, A, R, P, \gamma)$ be a finite MDP with $|S| < \infty$, $|A| < \infty$, bounded rewards $|R(s, a)| \leq R_{\max} \forall (s, a)$, and discount factor $\gamma \in [0, 1)$. In this section, we will first explore an alternative proof approach for the value iteration algorithm, then we cover policy iteration which is discussed in the class more precisely.

1.1 Positive Rewards

Assume $R(s, a) \geq 0$ for all s, a .

1. Derive an upper bound for the optimal k -step value function V_k^* . The optimal k -step value function satisfies:

Answer:

$$V_k^*(s) = \max_a \left[R(s, a) + \gamma \sum_{s'} P(s'|s, a) V_{k-1}^*(s') \right]$$

Given $R(s, a) \geq 0$, we prove by induction:

- Base case ($k = 0$): $V_0^*(s) = 0 \leq \frac{R_{\max}}{1-\gamma}$
- Inductive step: Assume $V_{k-1}^*(s) \leq \frac{R_{\max}}{1-\gamma}$, then

$$V_k^*(s) \leq R_{\max} + \gamma \cdot \frac{R_{\max}}{1-\gamma} = \frac{R_{\max}}{1-\gamma}$$

Thus, $V_k^*(s) \leq \frac{R_{\max}}{1-\gamma}$ for all k, s . □

2. Prove V_k^* is non-decreasing in k . Giving a policy π such that:

$$V_{k+1}^\pi \geq V_k^*.$$

Use this to show convergence of Value Iteration to a solution satisfying the Bellman equation.

Answer:

For any k , define policy π following the k -step optimal policy for first k steps:

$$V_{k+1}^\pi(s) = R(s, \pi(s)) + \gamma \sum_{s'} P(s'|s, \pi(s)) V_k^*(s') \geq V_k^*(s)$$

By Monotone Convergence Theorem, $V_k^* \rightarrow V^*$ satisfying $V^* = BV^*$. □

3. By taking the limit in the Bellman equation, prove that the V^* is optimal.

Answer:

Taking $k \rightarrow \infty$:

$$V^*(s) = \max_a \left[R(s, a) + \gamma \sum_{s'} P(s'|s, a) V^*(s') \right]$$

By Banach's Fixed-Point Theorem, V^* is unique and optimal. □

1.2 General Rewards

Remove the non-negativity constraint on $R(s, a)$. Assume no terminating states exist. Consider a new MDP defined by adding a constant reward r_0 to all rewards of the current MDP. That is, for all (s, a) , the new reward is:

$$\hat{R}(s, a) = R(s, a) + r_0$$

4. By deriving the optimal action and V_k^* in terms of the original MDPs values and r_0 , show that Value Iteration still converges to the optimal value function V^* (and optimal policy) of the original MDP even if rewards are negative. Also compute the new value V^* .

Answer:

For $\hat{R}(s, a) = R(s, a) + r_0$:

$$\hat{V}_k^*(s) = V_k^*(s) + \frac{r_0}{1 - \gamma}$$

The Bellman equation becomes:

$$\hat{V}_k^*(s) = \max_a \left[R(s, a) + r_0 + \gamma \sum_{s'} P(s'|s, a) \hat{V}_{k-1}^*(s') \right]$$

Subtracting $\frac{r_0}{1-\gamma}$ recovers original Bellman equation. □

5. Why is it necessary to assume the absence of a terminating state? Try to explain with a counterexample.

Answer:

Counterexample: If s_{end} exists with $R(s_{\text{end}}, a) = 0$, adding $r_0 > 0$ makes agents avoid termination to collect r_0 indefinitely. □

1.3 Policy Turn

In this part we want to dive into the mathematical proof of policy iteration.

6. Let π_k be the policy at iteration k . Prove the following:

$$V^{\pi_{k+1}}(s) \geq V^{\pi_k}(s) \quad \forall s \in S,$$

with strict inequality for at least one state unless π_k is already optimal. Use the definition of the greedy policy and explain why policy improvement leads to a better or equal value function.

Answer:

For greedy policy π_{k+1} :

$$V^{\pi_{k+1}}(s) \geq Q^{\pi_k}(s, \pi_{k+1}(s)) \geq V^{\pi_k}(s)$$

Strict inequality holds unless π_k is optimal. □

7. Prove that Policy Iteration always converges to the optimal policy in a finite MDP. Specifically, show that after a finite number of policy evaluations and improvements, the algorithm reaches a policy π^* that satisfies the Bellman optimality equation. You may use theorems discussed in class, but if a result was not proven, please provide a full justification.

Answer:

Finite MDPs have $|\mathcal{A}|^{|\mathcal{S}|}$ deterministic policies. Each iteration strictly improves V^π until optimal. $\square$

8. Prove that Value Iteration and Policy Iteration both converge to the same optimal value function V^* , even if the policies may differ. How the policies are still optimal despite possible differences?

Answer:

Both algorithms solve $V = BV$, which has unique fixed point V^* . Optimal policies may differ but satisfy $V^\pi = V^*$. $\square$

9. Compare and contrast the computational cost of one step of Policy Iteration (i.e., full Policy Evaluation + Policy Improvement) versus one iteration of Value Iteration.

Answer:

- Policy Iteration: $O(|\mathcal{S}|^3)$ (matrix inversion) + $O(|\mathcal{S}|^2|\mathcal{A}|)$ (improvement)
- Value Iteration: $O(|\mathcal{S}|^2|\mathcal{A}|)$ per iteration

10. In the context of a (MDP) with an infinite horizon, when the discount factor $\gamma = 1$, analyze how both Value Iteration and Policy Iteration behave.

Answer:

- Value Iteration: Diverges for infinite-horizon MDPs
- Policy Iteration: Fails as V^π becomes unbounded

Require $\gamma < 1$ for convergence. $\square$

2 Bellman or Bellwoman

[1] Recall that a value function is a $|S|$ -dimensional vector where $|S|$ is the number of states of the MDP. When we use the term V in these expressions as an arbitrary value function, we mean that V is an arbitrary $|S|$ -dimensional vector which need not be aligned with the definition of the MDP at all. On the other hand, V^π is a value function that is achieved by some policy π in the MDP. For example, say the MDP has 2 states and only negative immediate rewards. $V = [1, 1]$ would be a valid choice for V even though this value function can never be achieved by any policy π , but we can never have a $V^\pi = [1, 1]$. This distinction between V and V^π is important for this question and more broadly in reinforcement learning.

2.1 Bellman Operators

In the first part of this problem, we will explore some general and useful properties of the Bellman backup operator. We know that the Bellman backup operator B , defined below, is a contraction with the fixed point as V^* , the optimal value function of the MDP. The symbols have their usual meanings. γ is the discount factor and $0 \leq \gamma < 1$. In all parts, $\|v\| = \max_s |v(s)|$ is the infinity norm of the vector.

$$(BV)(s) = \max_a \left[r(s, a) + \gamma \sum_{s' \in S} p(s'|s, a) V(s') \right]$$

We also saw the contraction operator B^π with the fixed point V^π , which is the Bellman backup operator for a particular policy given below:

$$(B^\pi V)(s) = r(s, \pi(s)) + \gamma \sum_{s' \in S} p(s'|s, \pi(s)) V(s')$$

In this case, we'll assume π is deterministic, but it doesn't have to be in general. You have seen that $\|BV - BV'\| \leq \gamma \|V - V'\|$ for two arbitrary value functions V and V' .

1. Show that the analogous inequality, $\|B^\pi V - B^\pi V'\| \leq \gamma \|V - V'\|$, holds.

Answer:

For any policy π , B^π is a γ -contraction in $\|\cdot\|_\infty$.

Proof: Let $\delta = \|V - V'\|_\infty$. For any s :

$$|B^\pi V(s) - B^\pi V'(s)| \leq \gamma \sum_{s'} P(s'|s, \pi(s)) |V(s') - V'(s')| \leq \gamma \delta$$

Taking max over s gives $\|B^\pi V - B^\pi V'\|_\infty \leq \gamma \|V - V'\|_\infty$. $\square$

2. Prove that the fixed point for B^π is unique. Recall that the fixed point is defined as V satisfying $V = B^\pi V$. You may assume that a fixed point exists.

Answer:

Banach Fixed-Point Theorem guarantees unique fixed point for contractions.

Proof: From (1), B^π is a contraction. Since $V^\pi = B^\pi V^\pi$ exists, uniqueness follows. $\square$

3. Suppose that V and V' are vectors satisfying $V(s) \leq V'(s)$ for all s . Show that $B^\pi V(s) \leq B^\pi V'(s)$ for all s . *Note: all of these inequalities are elementwise.*

Answer:

Linear operators preserve elementwise inequalities.

Proof: For all s :

$$B^\pi V(s) = r(s, \pi(s)) + \gamma \sum_{s'} P(s'|s, \pi(s)) V(s') \leq r(s, \pi(s)) + \gamma \sum_{s'} P(s'|s, \pi(s)) V'(s') = B^\pi V'(s)$$

□

2.2 Bellman Residuals

We can extract a greedy policy π from an arbitrary value function V using the equation below:

$$\pi(s) = \arg \max_a \left[r(s, a) + \gamma \sum_{s' \in S} p(s'|s, a) V(s') \right]$$

It is often helpful to know what the performance will be if we extract a greedy policy from an arbitrary value function. To see this, we introduce the notion of a Bellman residual.

Define the Bellman residual to be $(BV - V)$ and the Bellman error magnitude to be $\|BV - V\|$.

4. For what value function V does the Bellman error magnitude $\|BV - V\|$ equal 0? Why?

Answer:

$\|BV - V\| = 0 \iff V = V^*$. Follows from Bellman optimality: $V^* = BV^*$. □

5. Prove the following statements for an arbitrary value function V and any policy π .

$$\begin{aligned} \|V - V^\pi\| &\leq \frac{\|V - B^\pi V\|}{1 - \gamma} \\ \|V - V^*\| &\leq \frac{\|V - BV\|}{1 - \gamma} \end{aligned}$$

Answer:

Triangle inequality and contraction properties.

Proof: Let $\delta = \|V - B^\pi V\|$. Then:

$$\|V - V^\pi\| \leq \sum_{k=0}^{\infty} \gamma^k \|B^\pi V - V\| = \frac{\delta}{1 - \gamma}$$

Similarly for V^* using B 's contraction. □

6. Let V be an arbitrary value function and π be the greedy policy extracted from V . Let $\varepsilon = \|BV - V\|$ be the Bellman error magnitude for V . Prove the following for any state s .

$$V^\pi(s) \geq V^*(s) - \frac{2\varepsilon}{1 - \gamma}$$

Answer:

From (5), $\|V - V^*\| \leq \frac{\varepsilon}{1-\gamma}$. Then:

$$V^\pi(s) \geq V(s) - \frac{\varepsilon}{1-\gamma} \geq V^*(s) - \frac{2\varepsilon}{1-\gamma}$$

□

7. Give an example real-world application or domain where having a lower bound on $V^\pi(s)$ would be useful.

Answer:

Medical treatment planning - guarantees policy performance stays within safe bounds relative to optimal.

8. Suppose we have another value function V' and extract its greedy policy π' . $\|BV' - V'\| = \varepsilon = \|BV - V\|$. Does the above lower bound imply that $V^\pi(s) = V^{\pi'}(s)$ at any s ?

Answer:

No. Different V, V' can yield $\pi \neq \pi'$ while having same ε . Example: $V = V^* + \epsilon$, $V' = V^* - \epsilon$.

Say $V \leq V'$ if $\forall s, V(s) \leq V'(s)$.

What if our algorithm returns a V that satisfies $V^* \leq V$? I.e., it returns a value function that is better than the optimal value function of the MDP. Once again, remember that V can be any vector, not necessarily achievable in the MDP, but we would still like to bound the performance of V^π where π is extracted from said V . We will show that if this condition is met, then we can achieve an even tighter bound on policy performance.

9. Using the same notation and setup as part 5, if $V^* \leq V$, show the following holds for any state s . Recall that for all π , $V^\pi \leq V^*$ (why?)

$$V^\pi(s) \geq V^*(s) - \frac{\varepsilon}{1-\gamma}$$

Answer:

Proof: From $V^\pi \leq V^* \leq V$ and (5):

$$V^\pi(s) \geq V(s) - \frac{\varepsilon}{1-\gamma} \geq V^*(s) - \frac{\varepsilon}{1-\gamma}$$

□

Intuition: A useful way to interpret the results from parts (8) and (9) is based on the observation that a constant immediate reward of r at every time-step leads to an overall discounted reward of

$$r + \gamma r + \gamma^2 r + \dots = \frac{r}{1-\gamma}$$

Thus, the above results say that a state value function V with Bellman error magnitude ε yields a greedy policy whose reward per step (on average), differs from optimal by at most 2ε . So, if we develop an algorithm that reduces the Bellman residual, we are also able to bound the performance of the policy extracted from the value function outputted by that algorithm, which is very useful!

10. It's not easy to show that the condition $V^* \leq V$ holds because we often don't know V^* of the MDP. Show that if $BV \leq V$ then $V^* \leq V$. Note that this sufficient condition is much easier to check and does not require knowledge of V^* .

Hint: Try to apply induction. What is $\lim_{n \rightarrow \infty} B^n V$?

Answer:

Proof: By induction, $B^n V \leq V$. Since $\lim_{n \rightarrow \infty} B^n V = V^*$, we get $V^* \leq V$. $\square$

11. (Bonus) It is possible to make the bounds from parts (9) and (10) tighter. Let V be an arbitrary value function and π be the greedy policy extracted from V . Let $\varepsilon = \|BV - V\|$ be the Bellman error magnitude for V . Prove the following for any state s :

$$V^\pi(s) \geq V^*(s) - \frac{2\gamma\varepsilon}{1-\gamma}$$

Further, if $V^* \leq V$, prove for any state s

$$V^\pi(s) \geq V^*(s) - \frac{\gamma\varepsilon}{1-\gamma}$$

Answer:

Proof Sketch: Iterate Bellman equation:

$$V^\pi \geq B^\pi V - \gamma\varepsilon \mathbf{1} \geq V^* - \frac{\gamma\varepsilon}{1-\gamma} \mathbf{1}$$

where $\mathbf{1}$ is all-ones vector. $\square$

References

- [1] Baesed on CS 234: Reinforcement Learning, Stanford University. Spring 2024.
- [2] [Cover image designed by freepik](#)